

Expressing Ideas Clearly and Coherently in Academic and Technical Forms

Producing academic and technical texts is different from producing casual writing or everyday speech. A well-written essay, research report, presentation, or technical manual must be clear, coherent, well-organized, and appropriate for its audience and purpose. It must also use precise vocabulary, integrate visuals where helpful, condense complex material without losing meaning, and follow ethical standards of attribution. These skills prepare students not only for higher education but also for workplaces where clear communication determines whether a project succeeds or fails. Expressing ideas well is not a talent reserved for gifted writers. It is a set of learnable practices that improve with attention and revision.

This reading material explains how to compose well-structured academic and technical texts with awareness of audience and purpose. It shows how to explain concepts using appropriate academic and technical vocabulary and multimodal representations. It demonstrates how to condense spoken, written, or multimodal texts into outlines, summaries, synopses, paraphrases, diagrams, and charts. Finally, it explains ethical considerations such as citation methods, AI attribution, and avoiding plagiarism when producing extended academic and technical texts.

Part 1: Composing Well-Structured Academic and Technical Texts

Good structure is not decoration. It is the visible form of clear thinking. Readers and listeners follow ideas more easily when the text signals where it is going, what matters most, and how the parts connect.

Common Academic and Technical Text Types

Text Type	Purpose	Typical Audience	Key Structural Features
Essay	To analyze, argue, or reflect on an idea	Teachers, fellow students, examiners	Introduction with thesis, body paragraphs with topic sentences and evidence, conclusion
Research report	To present investigation and findings	Academic readers, supervisors	Abstract, introduction, methodology, results, discussion, conclusion, references
Analysis report	To examine a problem and recommend action	Managers, clients, technical teams	Executive summary, background, analysis, findings, recommendations
Technical manual	To instruct users in operating or assembling	End users, technicians	Overview, safety notes, parts list, step-by-step procedures, troubleshooting
Presentation with visual aids	To inform or persuade a live audience	Classmates, panel, stakeholders	Opening hook, agenda, clearly sequenced sections, visuals, closing summary
Project proposal	To seek approval or funding	Administrators, sponsors	Problem, objectives, methodology, timeline, budget, expected outcomes

The IBC Structure: Introduction, Body, Conclusion

Nearly all academic and technical texts follow a three-part structure, though the labels change by genre.

Section	Function	What to Include
Introduction	Orients the reader	Context, purpose, scope, main claim or objective, roadmap
Body	Develops the content	Logical sections, evidence, examples, visuals, transitions
Conclusion	Closes the discussion	Summary of key points, implications, recommendation or final thought

Awareness of Audience and Purpose

Every structural decision reflects choices about audience and purpose.

Audience	What They Need	How to Structure Accordingly
Teacher or examiner	Evidence of reasoning and mastery	Clear thesis, logical progression, cited sources
Classmates	Clarity and relevance	Familiar examples, moderate detail, engaging opener
Technical specialists	Precision and data	Definitions, specifications, tables and figures
General public	Accessibility	Plain language, analogies, minimal jargon
Administrators or sponsors	Justification and feasibility	Problem statement, cost-benefit, timeline, recommendation
End users of a manual	Step-by-step clarity	Numbered steps, cautions, labeled diagrams

Paragraph Craft in Academic and Technical Writing

Element	Function	Example
Topic sentence	States the paragraph's main idea	"Regular sampling improves the reliability of water quality data."
Supporting evidence	Provides facts, data, or citations	"According to a 2023 study, weekly sampling reduced measurement error by 34%."
Explanation	Connects evidence to the claim	"This suggests that infrequent sampling misses short-term fluctuations."
Transition	Links to the next idea	"Beyond frequency, the method of collection also affects accuracy."
Concluding sentence (optional)	Reinforces or transitions	"Sampling design, therefore, is not a minor detail but a central concern."

Coherence and Cohesion

Coherence means the ideas fit together logically. **Cohesion** means the words and sentences connect smoothly through transitions, pronouns, and repeated key terms.

Tool	Purpose	Example
Transition words	Signal relationships	however, therefore, in addition, for example, in contrast
Pronouns and referencing	Avoid repetition and link ideas	"This finding supports the earlier claim."
Repetition of key terms	Maintain focus	Repeating "sampling method" rather than switching to "procedure" and "technique"

Tool	Purpose	Example
Parallel structure	Create rhythm and clarity	"The study measured depth, temperature, and turbidity."
Signposting	Guide the reader	"The next section examines three limitations."

Worked Example: Structuring a Short Analysis Report

Task: Analyze why Grade 11 students in a school are underutilizing the library.

Section	Content
Executive summary	Library visits declined 22% over one year; main causes are scheduling conflicts, outdated materials, and lack of awareness
Background	Brief context on the library's purpose and current usage data
Analysis	Three subsections: scheduling conflicts, collection relevance, communication gaps
Findings	Summary of the three main causes supported by survey data
Recommendations	Extend library hours, update the collection, and run an awareness campaign
References	Survey results, school records, and two studies on student library use

Checklist for Composing Academic and Technical Texts

Check	Question to Ask
Purpose	Is the goal clear within the first few sentences?
Audience	Is the vocabulary and level of detail appropriate?

Check	Question to Ask
Structure	Does each section serve a clear function?
Coherence	Do ideas follow logically from one to the next?
Cohesion	Are transitions and references used smoothly?
Evidence	Are claims supported by data, examples, or citations?
Conclusion	Does the ending summarize and point forward?

Part 2: Explaining Concepts Using Vocabulary and Multimodal Representations

Clear explanation requires two things: precise language and well-chosen visuals. Academic and technical texts are rarely text alone. They combine prose with diagrams, graphs, charts, and infographics to make complex ideas accessible.

Academic and Technical Vocabulary

Academic and technical language has recognizable features. It is precise, often formal, and relies on specific conventions.

Feature	Description	Example
Precision	Words are chosen for exact meaning	"The sample was heated to 60°C," not "heated a lot"
Nominalization	Actions expressed as nouns	"The evaporation of the liquid" instead of "the liquid evaporated"

Hedging	Careful qualification of claims	"The results suggest," "may indicate," "appears to be"
Passive voice	Emphasizes process or object over actor	"The solution was filtered" rather than "We filtered the solution"
Technical terms	Discipline-specific vocabulary	catalyst, coefficient, calibration, correlation
Defined terms	Terms introduced with definitions	"By 'sample size,' we mean the number of participants selected."
Objective tone	Avoids emotional or personal language	"The data indicate a decline" rather than "I feel the numbers are disappointing"

Avoiding Overly Informal or Vague Language

Vague or Informal	Academic or Technical Version
"A lot of people think..."	"A majority of respondents reported..."
"It's kind of important."	"It is a critical factor."
"The thing broke."	"The component failed."
"We did some tests."	"Three controlled trials were conducted."
"Pretty big difference."	"A statistically significant difference was observed."

Multimodal Representations: When to Use Each

Representation	Best For	Example
Diagram	Showing structure, parts, or relationships	Labeled drawing of a cell or circuit
Flowchart	Showing a process with steps and decisions	Laboratory procedure or workflow
Bar graph	Comparing categories	Enrollment per track
Line graph	Showing change over time	Monthly rainfall
Pie chart	Showing parts of a whole	Budget allocation
Scatter plot	Showing relationships between variables	Study time vs. test scores
Table	Presenting precise values	Experimental results with units
Infographic	Combining visuals and short text for general audiences	Public health awareness poster
Timeline	Showing chronological sequence	Project milestones
Concept map	Showing connections among ideas	Themes in a literary work

Principles of Effective Visual Design

Principle	Explanation
Relevance	The visual must directly support the text's point
Simplicity	Remove unnecessary decoration

Principle	Explanation
Labeling	Every axis, symbol, and component must be labeled
Consistency	Use the same colors, fonts, and units throughout
Scale accuracy	Axes should not be truncated to exaggerate differences
Accessibility	Use color contrast and avoid relying on color alone
Citation	Credit the source of data or images

Worked Example: Explaining a Concept with Text and Visuals

Concept: The water cycle

Prose explanation: Water moves continuously between the atmosphere, land, and oceans through evaporation, condensation, precipitation, and collection. Solar energy drives evaporation from bodies of water and soil. Water vapor rises, cools, and condenses into clouds. When droplets become heavy enough, they fall as precipitation. Water then collects in rivers, lakes, and underground reservoirs, and the cycle repeats.

Visual accompaniment: A labeled diagram showing the ocean, sun, evaporation arrows, cloud formation, precipitation, and runoff returning to the ocean.

Why both are needed: The prose explains causes and sequences; the diagram shows spatial relationships and simultaneous processes that would be tedious to describe in words alone.

Integrating Visuals with Text

Technique	How It Works	Example
Reference in text	Direct the reader to the visual	"As shown in Figure 1, precipitation peaks in July."

Technique	How It Works	Example
Caption	Brief explanation below the visual	"Figure 1. Average monthly rainfall, 2020–2023."
Annotation	Notes directly on the visual	Labels on a diagram indicating parts
Cross-comparison	Compare two visuals side by side	Before-and-after charts
Highlighting	Emphasize a key value or trend	Bold line showing a target threshold

Part 3: Condensing Texts into Outlines, Summaries, and Visuals

Academic and technical work often requires reducing a longer text to its essentials. Condensing is not simply cutting words. It is identifying what matters most and representing it accurately in a shorter form.

Common Condensing Forms

Form	Description	Length Relative to Source	Keeps
Outline	Hierarchical list of main ideas and supporting points	Very short	Structure and hierarchy
Summary	Brief restatement of main ideas in prose	10–25%	Main ideas and key evidence

Synopsis	Condensed overview, often of a narrative or report	5–10%	Sequence and essential events or findings
Paraphrase	Restatement of a specific passage in one's own words	Similar to original passage	Full meaning of the passage
Abstract	Formal summary at the start of a research paper	150–250 words	Purpose, method, results, conclusion
Diagram or chart	Visual condensation of relationships or data	Minimal text	Patterns, structure, or comparisons
Precis	Careful, compact restatement preserving proportion	About one-third	Argument and emphasis

Outline: Structuring Main and Supporting Ideas

An outline uses headings and subheadings to show hierarchy.

Example Outline: Water Quality Report

- I. Introduction
 - A. Purpose of the study
 - B. Scope and location
- II. Methodology
 - A. Sampling sites
 - B. Collection and testing procedures
- III. Results
 - A. pH levels
 - B. Turbidity readings
 - C. Bacterial content
- IV. Discussion
 - A. Comparison with standards
 - B. Possible sources of contamination
- V. Recommendations

Summary: Restating Main Ideas

A summary answers the question: "What is this text mainly saying?" It omits examples, repetitions, and minor details.

Guidelines for Summarizing

Guideline	Explanation
Identify the main idea	Locate the central claim or purpose
Include key supporting points	Retain evidence that directly supports the main idea
Omit examples and repetition	Cut illustrations and restatements
Use your own words	Do not copy sentences verbatim
Preserve proportion	Give more space to important ideas, less to minor ones
Cite the source	Always indicate where the summary came from

Example Summary

Original passage (approximately 120 words): A study conducted over six months in three coastal barangays found that water quality varied significantly with rainfall. During dry months, bacterial contamination remained within safe limits at all three sites. After heavy rains, contamination levels rose sharply, particularly at the site nearest to an open drainage system. The researchers attributed this to surface runoff carrying waste into the water. They recommended regular testing during the rainy season and improved drainage infrastructure.

Summary (approximately 35 words): A six-month coastal water quality study found that contamination remained safe in dry months but rose sharply after heavy rains, especially near open drainage. Researchers recommended rainy-season testing and improved drainage.

Synopsis: A Broader Condensation

A synopsis is similar to a summary but often covers a larger work — a report, a study, or a narrative — in a shorter, more sweeping form. It emphasizes the arc or overall outcome rather than individual supporting points.

Example Synopsis of a Research Report

This report examines student library usage over one academic year, identifies three main causes of declining visits, and recommends extending hours, updating the collection, and improving communication.

Paraphrase: Restating a Specific Passage

A paraphrase restates a short passage in different words while preserving the full meaning. It is longer than a summary and closely follows the original's content.

Original: "The viscosity of the liquid increased as the temperature dropped, slowing its flow through the narrow tube."

Paraphrase: As the temperature fell, the liquid became thicker, causing it to move more slowly through the narrow tube.

Condensing into Diagrams and Charts

Sometimes the best condensation is visual rather than verbal. A paragraph describing a process can become a flowchart. A page of statistics can become a chart.

Original Form

Condensed Visual Form

Paragraph describing steps

Flowchart

Data table with 20 rows

Bar graph or line graph

Report section on causes and effects

Cause-and-effect diagram

Chapter on relationships among themes

Concept map

Original Form

Condensed Visual Form

Timeline in prose

Horizontal timeline

Comparison of two options

Side-by-side table

Worked Example: Condensing a Written Procedure into a Flowchart

Original text: To test the sample, first calibrate the meter using the standard solution. If the reading is within range, proceed to measurement. If it is out of range, recalibrate and repeat the check. Once calibrated, immerse the probe in the sample, wait thirty seconds, and record the reading. Repeat for each sample and rinse the probe between readings.

Condensed flowchart steps:

1. Calibrate meter
2. Check reading
3. Decision: within range?
 - Yes → proceed to measurement
 - No → recalibrate and return to step 2
4. Immerse probe in sample
5. Wait 30 seconds
6. Record reading
7. Rinse probe
8. Repeat for next sample

Checklist for Condensing Accurately

Check	Question
Accuracy	Does the condensed version preserve the original meaning?
Completeness	Are the main ideas all present?
Proportion	Are important ideas given more emphasis than minor ones?
Originality	Is the wording your own, except for quoted phrases?
Citation	Is the source clearly identified?
Clarity	Would a reader unfamiliar with the original understand it?

Part 4: Ethical Considerations in Producing Academic and Technical Texts

Producing extended academic and technical texts carries ethical responsibilities. Writers must credit sources, represent others' ideas fairly, avoid plagiarism in all its forms, and be transparent about the use of AI tools.

Why Ethics Matter

Reason	Explanation
Respect for intellectual work	Writers and researchers deserve credit for their labor
Academic integrity	Honesty is a core value of scholarship
Credibility	Ethical work can be trusted; unethical work cannot

Reason	Explanation
Legal compliance	Copyright and fair use laws apply
Professional standards	Workplaces expect honesty in reports and proposals

Citation Methods

Citations give credit to the original source and allow readers to verify claims. Different disciplines use different styles.

Style	Common In	In-Text Example	Reference List Example
APA (7th ed.)	Social sciences, education, psychology	(Santos, 2023, p. 45)	Santos, M. (2023). <i>Title of work</i> . Publisher.
MLA (9th ed.)	Humanities, literature	(Santos 45)	Santos, Maria. <i>Title of Work</i> . Publisher, 2023.
Chicago	History, some humanities	Footnote or endnote	Santos, Maria. <i>Title of Work</i> . Publisher, 2023.
IEEE	Engineering, computer science	[1]	[1] M. Santos, <i>Title of Work</i> . City: Publisher, 2023.

When to Cite

Situation	Cite?
Direct quotation	Yes, with quotation marks and citation
Paraphrase of a specific idea	Yes

Situation	Cite?
Summary of another's work	Yes
Statistical data from a source	Yes
Common knowledge (e.g., "The Philippines has 7,641 islands")	Usually no
Your own original analysis	No

Quoting, Paraphrasing, and Summarizing

Technique	What It Does	When to Use
Direct quotation	Reproduces exact words	When the original phrasing is distinctive or authoritative
Paraphrase	Restates a passage in your own words	When you need to convey specific content without quoting
Summary	Condenses the main ideas	When you need an overview of a longer source

All three require citation. Quoting also requires quotation marks and page numbers where applicable.

Forms of Plagiarism

Type	Description	Example
Direct plagiarism	Copying words without quotation or citation	Pasting a paragraph from a website
Mosaic plagiarism	Mixing copied phrases with your	Changing a few words but keeping

Type	Description	Example
Self-plagiarism	own words without citation	the structure
	Reusing your own previous work without disclosure	Submitting the same essay for two subjects
Accidental plagiarism	Forgetting to cite or misattributing	Losing track of a source
Idea plagiarism	Presenting another's original idea as your own	Using a theory without crediting its author

AI Attribution and Declaration

AI tools can assist with brainstorming, drafting, editing, and formatting, but their use must be transparent.

Use of AI	Acceptable?	How to Disclose
Brainstorming ideas	Usually yes	Mention in a note or methods section if required
Generating an outline	Depends on institutional policy	Disclose in an acknowledgments note if used
Drafting full sections	Usually no for assessed work	Not permitted in most academic contexts without permission
Editing grammar and style	Often permitted	Check policy; disclose if required
Generating citations	Not reliable; verify manually	Never cite AI-generated references without verification

Use of AI	Acceptable?	How to Disclose
Translating text	Permitted with caution	Disclose if required; verify accuracy

A sample declaration: "The author used [AI tool] to generate an outline and to check grammar. All content, analysis, and sources were written and verified by the author."

Ethical Practices for Extended Texts

Practice	Explanation
Keep a research log	Record sources as you find them, not later
Quote accurately	Never alter a quotation without using brackets or ellipses
Paraphrase honestly	Do not simply replace words with synonyms
Cite consistently	Follow one style guide throughout
Acknowledge collaborators	Credit group members for their contributions
Disclose AI use	Follow your institution's policy exactly
Respect copyright	Do not reproduce large portions of copyrighted work without permission
Represent sources fairly	Do not distort another author's argument to make it easier to refute

Worked Example: Correcting an Ethical Error

Original source: "Urban heat islands can raise local temperatures by as much as 7°C compared to surrounding rural areas."

Unethical use: Urban heat islands raise temperatures by 7°C. *(No citation, overstated as universal.)*

Ethical use (quotation): According to Santos (2022), "urban heat islands can raise local temperatures by as much as 7°C compared to surrounding rural areas" (p. 112).

Ethical use (paraphrase): Santos (2022) reported that urban heat islands may increase local temperatures by up to 7°C above nearby rural areas.

Ethical use (summary): Research indicates that urban heat islands can significantly raise local temperatures relative to surrounding rural regions (Santos, 2022).

Key Terms

Term	Definition
Academic text	A text that analyzes, argues, or reports knowledge for scholarly purposes
Technical text	A text that explains or instructs in a specific field
Coherence	The logical connection of ideas within a text
Cohesion	The use of linguistic devices to link sentences and paragraphs
Thesis statement	The central claim of an essay, usually in the introduction
Topic sentence	The sentence stating a paragraph's main idea
Multimodal representation	A visual form such as a diagram, graph, chart, or infographic
Outline	A hierarchical list of main and supporting ideas
Summary	A brief restatement of a text's main ideas
Synopsis	A condensed overview of a longer work
Paraphrase	A restatement of a passage in one's own words
Abstract	A formal summary at the beginning of a research paper
Citation	A reference to the source of information or ideas
Plagiarism	Presenting another's words or ideas as one's own
Mosaic plagiarism	Mixing copied phrases with original wording without citation

Term	Definition
Self-plagiarism	Reusing one's own prior work without disclosure
AI attribution	Disclosing the use of artificial intelligence tools in producing a text
Hedging	Using cautious language to qualify claims

Review Questions

1. Explain how audience and purpose influence the structure of an academic or technical text. Provide one example.
2. Differentiate coherence from cohesion. Provide one example of each.
3. Identify four features of academic and technical vocabulary and give an example of each.
4. Choose three multimodal representations and explain what each is best used for.
5. A paragraph describes a five-step laboratory procedure. What visual form would best condense it, and why?
6. Differentiate a summary from a synopsis and from a paraphrase.
7. List five guidelines for writing an accurate summary.
8. Identify three forms of plagiarism and describe how each can be avoided.
9. Study the source and student text below.

Source: "Regular physical activity improves cardiovascular health and reduces the risk of chronic disease."

Student text: Regular exercise makes your heart healthier and lowers the chance of getting sick.

- (a) Is the student text a quotation, paraphrase, or summary?
 - (b) What ethical requirement is missing?
 - (c) Rewrite the student text with proper attribution using APA style.
- 10.** Application question: You are asked to produce a research and analysis report on study habits among Grade 11 students, using survey data you collected. Describe how you would (a) structure the report, (b) use vocabulary and visuals to explain the findings, (c) condense your survey results into a chart and a summary, and (d) apply ethical considerations in citing sources and disclosing AI use if any.